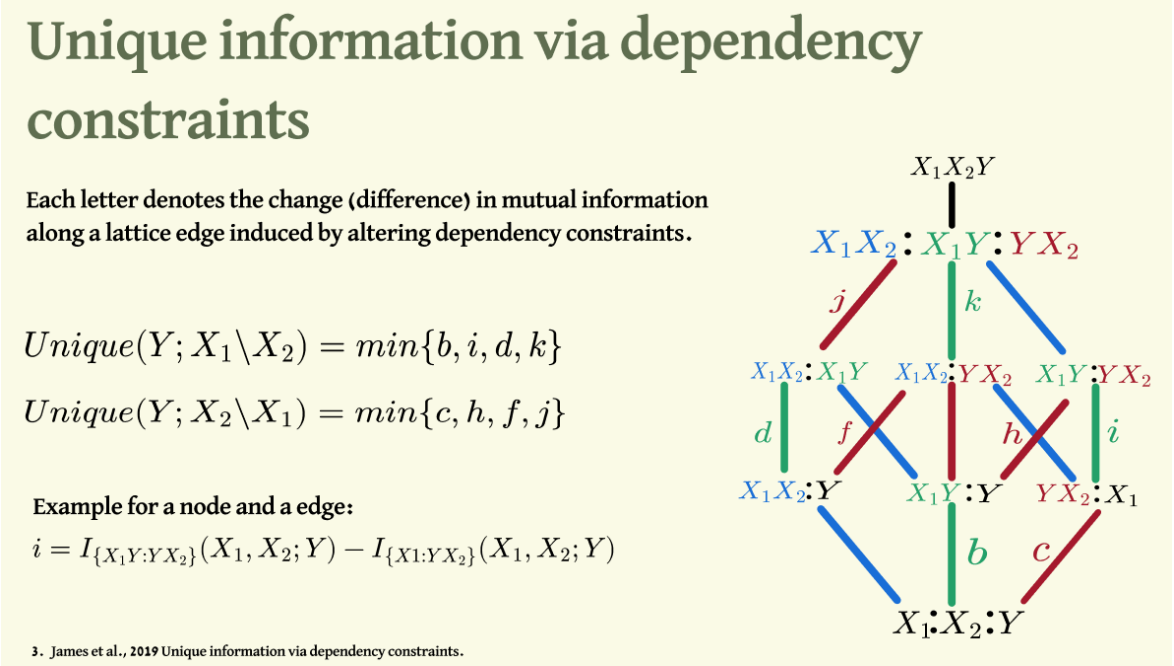
Progress

M7 and M8 in Idep:



In his paper “Idep Unique information via dependency constraints” James introduces the unique information calcualted as the least change in the joint mutual infomation between edges i.e the minimal difference between joint information to connected with same color nodes..
Or as in the image the minimum between edges. At the appendix of his paper he proves that for

$$Unique(Y; X_1 \backslash X_2) = min\{i, k\}$$

$$Unique(Y; X_2 \backslash X_1) = min\{j, h\}$$

Calculating when RVs are jointly gaussian:

In the paper “Exact Partial Information Decompositions for
Gaussian Systems Based on Dependency Constraints” the authors show a framework on how to calcualte the Idep when assuming the RVs are jointly gaussian and when we have the covariance.

Each edge corrspond to some contstraint where in the gaussain case the constraint is:

First we look at the precision matrix (the inverse of the covariace):

$$\Sigma^{-1} = K = \begin{pmatrix} K_{11} & K_{12} & K_{1T} \\ K_{21} & K_{22} & K_{2T} \\ K_{T1} & K_{T2} & K_{TT} \end{pmatrix}$$

Then we put zeroes at the entries matching the constraint. The zeroes represent conditional independence. For instance if $K_{12} = K_{21} = 0$ then: $X1 \perp\!\!\!\perp X2|T$

(The example will be the only important constraint other then the node which represents no constraints) We will go over it later on.

We will not go over all the nodes but only the improtant ones.

First we have sample covariance:

$$\Sigma = \begin{bmatrix} \Sigma_{00} & \Sigma_{01} & \Sigma_{02} \\ \Sigma_{01}^T & \Sigma_{11} & \Sigma_{12} \\ \Sigma_{02}^T & \Sigma_{12}^T & \Sigma_{22} \end{bmatrix}.$$

We can turn into:

$$\Sigma_Z = \begin{bmatrix} I_{n_0} & P & Q \\ P^T & I_{n_1} & R \\ Q^T & R^T & I_{n_2} \end{bmatrix}, \tag{32}$$

where the matrices P, Q, R are of size $n_0 \times n_1, n_0 \times n_2, n_1 \times n_2$, respectively, and are the cross-covariance (correlation) matrices between the three pairings of the three vectors $\mathbf{X}_0, \mathbf{X}_1, \mathbf{Y}$, and so

$$\mathbb{E}(\mathbf{X}_0\mathbf{X}_1^T) = P, \quad \mathbb{E}(\mathbf{X}_0\mathbf{Y}^T) = Q, \quad \mathbb{E}(\mathbf{X}_1\mathbf{Y}^T) = R, \tag{33}$$

defined on $\mathbb{R}^m$, where $m = n_0 + n_1 + n_2$.

Which we can create using the sample covariance and cholesky decomposition.

$$P = \left[\Sigma_{00}^{-\frac{1}{2}}\right]^T \Sigma_{01}\Sigma_{11}^{-\frac{1}{2}}, \quad Q = \left[\Sigma_{00}^{-\frac{1}{2}}\right]^T \Sigma_{02}\Sigma_{22}^{-\frac{1}{2}}, \quad R = \left[\Sigma_{11}^{-\frac{1}{2}}\right]^T \Sigma_{12}\Sigma_{22}^{-\frac{1}{2}}. \tag{A5}$$

Now we will focus on the two models that we need to calcualte Idep:

First:

Model M8

This is the regular model where we dont have any constraint and we calculate the mutual information using the regular formula for

$$I_{M8} = I((X_1, X_2), T)$$

Therefore:

$$I_{M8} = 0.5 * (log(|I - P.T@P|) - log(|\Sigma_Z|))$$

Because $\hat{\Sigma}$ is a sample covariance it is a wishart distribtuion $\hat{\Sigma} \sim W(\Sigma, n-1, p)$ with n-1 df and d as the feature dimension it has known bias correction.

> where $\psi(\cdot)$ is the **digamma function** (the logarithmic derivative of the Gamma function).
>
> To get the **unbiased estimate** of the log-determinant of the true covariance matrix Σ from your sample matrix $\hat{\Sigma}$, you apply the following correction:

$$\log|\Sigma|_{unbiased} = \log|\hat{\Sigma}| + \underbrace{d\log\left(\frac{N}{2}\right) - \sum_{i=1}^{d}\psi\left(\frac{N-i}{2}\right)}_{\text{Bias Correction Term}}$$

Where d is the dimnesion of the covariance matrix/block So for model M8 we dont have any problems of calculating the true value.

For now we will note the “Bias Correction Term” as $BC[(df, dimensions)]$

Model M7

Model M7 is under the assumption that $X1 \perp\!\!\!\perp X2|T$ and the oppiste,

Now in the second paper using the precision matrix they prove that the covariance will look like this:

$$\Sigma_7 = \begin{bmatrix} I_{n_0} & QR^T & Q \\ RQ^T & I_{n_1} & R \\ Q^T & R^T & I_{n_2} \end{bmatrix}.$$

That is because according to gaussian graphical models (GGM) the entries for where P is at in the precision matrix will be all zeros which symbolize the conditional independence.

The calcualtions for $I_{M7} = I((X_0, X_1), T)$ are:

$$I_{M7} = 0.5 * (log(|I - P.T@P|) - log(|\Sigma_7|))$$

That is where the problems begin:

1. Calculating the the right side in the difference **was** a problem but it is not a problem anymore. Why? Because under GGM there is a paper that proves that we can create a moment generating function that seperates the logdet into cliques and sperators where in our case T is the sperator and $X_0 - T$ AND $X_1 - T$ are cliques and the bias correction for that would be:
   (the paper called: “Exact formulas for the normalizing constants of Wishart distributions for graphical models”)

$$Bias(log(|\Sigma_{M7}|)) = BC[(df, d1 + dt)] + BC[(df, d2 + dt)] - BC[(df, dt)]$$

Which for simulations it works excpetionaly great.

The unsolved prbolem:

The left side in the difference. I shall call the “The Devil”

$$log(|I - P.T@P|)$$

Because $P = Q@R.T$ I am not able to derive a bias correction for it. Not analytical (Maybe there is but I can’t find it).

Parametric or a-parametric bias correction like Jackknife or paramteric bootstrapping are not good enough if $d \sim N$ which it is my case.

Back to Idep

Reminder:

$Unique(Y; X_1 \backslash X_2) = min\{i, k\}$ and $Unique(Y; X_2 \backslash X_1) = min\{j, h\}$

The calcuations:

$i = I_{M7} - I(X_2; T) = I_{M7}(T; X_1|X_2)$

$k = I_{M8} - I(X_1; T) = I_{M8}(T; X_2|X_1)$

$h = I_{M7} - I(X_1; T) = I_{M7}(T; X_2|X_1)$

$j = I_{M8} - I(X_1; T) = I_{M8}(T; X_2|X_1)$

Where the bi-variate mutual information it is easy to calcule with bias correction because it also a wishart matrix.

$$I(X1; T) = -0.5log(|I - Q.T@Q|) - BC[(df, d1 + dt)]$$

$$I(X2; T) = -0.5log(|I - R.T@R|) - BC[(df, d2 + dt)]$$

One more smaller problem:

If we will take the minimum of the nodes after bias correction it can be the flase for two reaasons:
1. If M7 is not fully corrected or corrected differently we might take the node related to min even though is not the corrected one.

2. We might take a minimum because of a small minus which is not corrected.

Suggested Solution:
1.Take the minimum before the bias correction and then correct only the one that is needed

2. **Report all nodes value and not just the minmum if you know they are both zero or both above zero than you can be sure that the unique is zero or above zero respectivly.**

What has been done so far:

1. Leave-one-out jackknife (Using with Oracle-Shrinkage aswell).
2. Parametric bootstrap (For unique information and for only “The Devil”).
3. Delta Method (Not sure I done it correctly).
4. Trying to guess a wishart bias correction.
5. Doint everything without whitening.

None of them really worked.

Jackknife only “The Devil” work the best.

More stuff to check

1. Jackknife with leave-d-out
2. Ledoit-Shrinkage
3.

$$\det\begin{pmatrix} I_{n_0} & QR^\top \\ RQ^\top & I_{n_1} \end{pmatrix} = \det\left(I_{n_0} - QR^\top RQ^\top\right) = \det\left(I_{n_1} - RQ^\top QR^\top\right)$$

The Schur complement commutes so it can help with dimension reduction.

More:

Optional Directions:

1. N >> p - BEST OPTION - Nothing can beat it.
2. N>p + analytical/resample bias correction to what we already have. The devil Term has smaller bias than the other terms.
3. Above we can see that the determinant of the the The Devil Term commutes. Therefore we can do dimension reduction to X0 or X1 or T and than to force N>>p. NOTE: We need to make sure it doesn't causes any other troubles.
4. Shrinkage to covariance. I am still no convinced that this would help.
5. Ridge Regression to covariance or predictors. I am still no convinced that this would help. Out of ridger regression we can find a model of coditional dependence. T∗B = X1 OR T∗B = X2.

If we do ridge regression Tal think MI wouldn't be biased.

1. Maybe we can build M7 covariance differently from how Ince builds it.
2. T is an R2 vector. Is there a quantile difference between 1 dimension than 2 dimensions.

Bias solution:

So far we couldn't solve analytically "The devil Term". Moreover bootstrap and jackknife didn't help as well. But using permutation to calculate the bias does help to find **lower bound** to unique information either we do it on the unique information itself or "The Devil Term" only and the rest components using analytical bias correction.

Lower bound is good enough to prove that unique information is above zero.

Now we can go on.

BUT - The problem that caused us to stop using IDEP:

Updates 08.05:

In order to be sure we can continue to real models and real brain activity we need to know that Idep doesn't fall to 3 supression examples. (You can skip to Evil-Twin Example)

Let (again) X_1, X_2 and T be sources and targer repsectivaly. We say X_2 is a supressor if we have the next 3:

1. $Unique1 >, Red = 0, Syn > 0$

2. $Unique1 > 0, Red > 0, Syn > 0$
3. $Unique1 = 0, Red = 0, Syn > 0$

In all cases $Unique2 = 0$. This means X_2 knows nothing about T that X_1 doesn't know. and it erases NOISE that helps X_1 predict T better. in the third case X_1 can be a suprresor for X_2 as well.
For examples 1 and 3 Idep solves easily.

Suppresion Paradox:

For example 2 Idep doesn't solve constantly and it depends on the Unique intensity in relation to Red intensity from simulation I ran. That happens (According to chat GPT) because Idep tests only pairwise dependencies in a Gaussian assumption (Not sure if it is outside the Gaussian world as well). Therefore when this happens it assign $Red = 0$ (I still don't have analytical proof) . Which is wrong, but maybe it is not so bad for out end goal. The problem is that on simulations I ran when the intensity of red and unique is the same Idep assigns $Unique2 > 0$ or $Unique2 \approx 0$ Which is not true and this is the BIGGER problem. We cannot have a definition assign unique information when is none.

The function I have used:

```python
def unq2_zero(rng, n, p, noise_std):
    """
    Intended theoretical structure:

        Y  = R + U + eps_y
        X1 = R + U + N
        X2 = R     + N

    X2 has:
        - redundancy through R
        - no private signal source about Y
        - synergistic/suppressor role through shared nuisance N

    So we expect:
        unq2 = 0
        redundancy > 0
        unq1 > 0
        synergy > 0
    under MMI-like Gaussian PID.
    """

    R = rng.standard_normal((n, p))   # redundant signal
    U = rng.standard_normal((n, p))   # unique-to-X1 signal
    N = noise_std * rng.standard_normal((n, p))  # shared suppressor noise

    eps_y = noise_std * rng.standard_normal((n, p))

    u_scale = 5.0
    r_scale = 0.7
    noise_scale = 2.0


    y = r_scale*R + u_scale*U + eps_y
    X_M1 = r_scale*R + u_scale*U + noise_scale*N
    X_M2 = r_scale*R + noise_scale*N

    print(f"Using scalses: R={r_scale}, U={u_scale}, N={noise_scale}, eps_y={noise_std}")
    return X_M1, X_M2, y
```

I just realized that this might even a bigger problem than we think (10.05.26) we (the team that think there is unique information) give us a sure win. Because we allow Unique=0 → Redundancy=0 but Redundancy=0 won't apply that Unique=0. This is true only we assume Gaussian distribution. Outside of the Gaussian world this might no happen.

Note (Maybe a Saver!):

I am not sure this a problem for us. As we look at commonality analysis results we can see that $R^2 \approx 10^-3$ which is very low. I think we should look for an example that is more realistic and has similar properties.

What to do next?

1. Understand if this is a problem we need to be concerd about.
2. Maybe we need to do some feature manipulation like PCA or CCA before computing Idep.
3. Another PID definition. ChatGPT says that a definition that repsects blackwell property can be helpful though the problem with definitions that respect blackwelian property is:

   BROJA is an upper bound and it's optimization problem and if we use Kolchinsky definition we don't any T-multivariate implementation. And hared exclusions (Ehrlich et al.) definition outputs negative atoms.

   One more suggestion is paper about Secret Key agreement can be helpfull as well. **(Right now there isn't a definition that can answer this perticular problem and not ruin something else).**

Let T be the target and let X_1, X_2 be two sources. Treat each source as a **channel from the target**:

$$\kappa_i(x_i \mid t) = P(X_i = x_i \mid T = t), \qquad i = 1, 2.$$

Blackwell order

We say that X_2 is **Blackwell-inferior** to X_1 about T, written

$$X_2 \preceq_T X_1,$$

if there exists a Markov kernel $K(x_2 \mid x_1)$, independent of T, such that

$$P(X_2 = x_2 \mid T = t) = \sum_{x_1} K(x_2 \mid x_1) P(X_1 = x_1 \mid T = t)$$

for every t, x_2.

Equivalently,

$$\kappa_2 = \kappa_1 K.$$

So X_2 can be generated from X_1 by applying a target-independent noisy channel.

In Markov-chain notation, this means there exists a random variable $\tilde{X}_2$ with the same joint law with T as X_2, such that

$$T \to X_1 \to \tilde{X}_2$$

is a Markov chain.

A **Markov kernel** is the mathematical object that represents a possibly noisy map from one space to another.

Let $(\mathcal{X}_1, \mathcal{F}_1)$ and $(\mathcal{X}_2, \mathcal{F}_2)$ be measurable spaces. A Markov kernel from $\mathcal{X}_1$ to $\mathcal{X}_2$ is a map

$$K : \mathcal{X}_1 \times \mathcal{F}_2 \to [0, 1]$$

such that:

For every fixed $x_1 \in \mathcal{X}_1$,

$$A \mapsto K(x_1, A)$$

is a probability measure on $(\mathcal{X}_2, \mathcal{F}_2)$.

And for every fixed measurable set $A \in \mathcal{F}_2$,

$$x_1 \mapsto K(x_1, A)$$

is a measurable function.

So $K(x_1, A)$ means:

$$K(x_1, A) = \Pr(\tilde{X}_2 \in A \mid X_1 = x_1).$$

Evil Twin - (Is it possible?)

We want to see if under the assumption of jointly Gaussian random variables we can construct two different examples (story wise) that have different PID values but the same covariance.

Example:

Probably the most important example: In one story X_2 has no unique information and in the second story it does have. Can it happen?

Example 1:

STORY 1:

$$X_1 = R + U_1 + \tilde{N} \mid X_2 = R + U_2 + \tilde{N} \mid T = R + U_1 + U_2 + N$$

$$p_1 = COV(X_1, X_2) = VAR(R) + VAR(\tilde{N})$$

$$q_1 = COV(X_1, T) = VAR(R) + VAR(U_1)$$

$$r_1 = COV(X_2, T) = VAR(R) + VAR(U_2)$$

STORY 2:

$$X_1 = R + U_1 + \tilde{N} \mid X_2 = R + \tilde{N} \mid T = R + U_1 + U_2 + N$$

$$p_2 = COV(X_1, X_2) = VAR(R) + VAR(\tilde{N})$$

$$q_2 = COV(X_1, T) = VAR(R) + VAR(U_1)$$
$$r_2 = COV(X_2, T) = VAR(R)$$

As we can see $p_1 = p_2$ and $q_1 = q_2$. The only thing that changes is r.

$R, U_1, U_2, \tilde{N}, N$ are all mutually independent.

but in order to have r_1=r_2 we get the following result:

$$VAR(R) + VAR(U_2) = VAR(R)$$
$$\downarrow$$
$$VAR(U_2) = 0$$

Therefore it can only happen when U_2 is a constant.

Note:

This is only example maybe we can find another.

Example 2:

STORY 1:

$$X_1 = R + U_1 + N \mid X_2 = R + U_2 + N \mid T = R + U_1 + U_2 + N_t$$
$$p_1 = COV(X_1, X_2) = VAR(R) + VAR(N)$$
$$q_1 = COV(X_1, T) = VAR(R) + VAR(U_1)$$
$$r_1 = COV(X_2, T) = VAR(R) + VAR(U_2)$$

STORY 2:

$$X_1 = \tilde{R} + \tilde{U_1} + \tilde{N} \mid X_2 = \tilde{R} + \tilde{N} \mid T = \tilde{R} + \tilde{U_1} + \tilde{U_2} + \tilde{N}_t$$
$$p_2 = COV(X_1, X_2) = VAR(\tilde{R}) + VAR(\tilde{N})$$
$$q_2 = COV(X_1, T) = VAR(\tilde{R}) + VAR(\tilde{U_1})$$
$$r_2 = COV(X_2, T) = VAR(\tilde{R})$$

Now it is not immediate that $p_1 = p_2$ and $q_1 = q_2$!

A good Evil Twin Example:

Evil_Twin___Covariance

20.05.26

After finding the Evil Twin example where we get the same covariance matrix for two scenarios: where in one case X_2 has unique information but in case 2 X_2 has zero unique information. I found out the Idep can't distinguish between the two and it chooses the less conservative - meaning it chooses the option where Unique2>0 which is less preferable for because we want a definition that at least in that case chooses the option where Unique2=0.

For that I have come to the realization that in order for X_2 to have Unique2=0 it is happens if and only if it is blackwellian inferior to X_1 with respect to T. James in his paper Idep says that his definition doesn't respect that property (A bug or a feature?). One definition that does choose to more conservative side is the BROJA in particular the Gaussian algorithm proposed by Venkatesh (2023). The problem with that definition that is usually (out side of the blackwellian inferior) an upper bound to the unique information and as a result we cannot asses well the magnitude of the true unique information. Venkatesh propsed another measure (That have to PID related but it is a lower bound to the unique information of **BROJA)** called: "The weighted output deficiency" which also has the blackwellian property (For that he derived a unique information definition called delta-PID) . We don't know if it is a lower bound to true unique information or a much tighter upper bound but this is good direction to check.

To Do:

- [x] ~~Check delta-PID on evil twin~~
- [x] ~~Read more about delta-PID~~
- [x] ~~See if we can find some upper/lower bound properties for delta-PID~~
- [x] ~~To see why Idep doesn't respect balckwell property~~
- [x] ~~To ask GPT to do a report about the blackwellian property in relation to PID~~
- [x] ~~To see if there is syn=0 and BROJA optimization is gurenteed. Not to find it but there exists one like that. - There isn't~~
- [x] ~~Run DeltaVsBROJAVsIdep with mi ground truth outside of algorithm.~~
- [x] ~~learn to work with CODEX~~
- [x] ~~To see if blackwell property can make a small searching space for lower bound for unique information.~~
- [x] ~~Check Idep's code written by Ince. (Do it with an agent)~~

Before choosing Blackwell property we need to choose the right story to tell.
Do we choose to believe dependencies or decision theoretic explanation. After doing that we can decide how to move on.

- [x] ~~Make a report about the two different philosophies of dependencies or decision theoretic.~~

Condition to choose the right PID definition:

Let's define number of tests for minimal requirments to choose the right definition:

1. Suppresion effect (All three examples)
2. Jointly Gaussian where there is unqiue>0 for both sources and it will find this.
3. Evil Twin - be more conservative. Choose unique = 0 for covariance.

4. If Correlation between X1 to X2 than R=0

Reorder this part.

Ince's counter example for BROJA

Ince's ReducedOR Example: Criticism of BROJA

Summary

Ince's **ReducedOR** example criticizes the BROJA definition of unique and redundant information. The example shows that preserving only the source-target marginals, $P(X_1, S)$ and $P(X_2, S)$, may be insufficient for deciding whether information is truly redundant or unique.

Setup

The system is similar to an OR gate:

$$S = X_1 \vee X_2$$

However, the event $(X_1, X_2) = (1, 1)$ never occurs.

Distribution

Event	Probability
$P(0,0,0)$	$\frac{1}{2}$
$P(1,0,1)$	$\frac{1}{4}$
$P(0,1,1)$	$\frac{1}{4}$

So the only possible realizations are:

- $X_1 = 0, X_2 = 0 \Rightarrow S = 0$
- $X_1 = 1, X_2 = 0 \Rightarrow S = 1$
- $X_1 = 0, X_2 = 1 \Rightarrow S = 1$

Why Both Predictors Are Informative

Both X_1 and X_2 contain information about S:

- If $X_1 = 1$, then $S = 1$
- If $X_2 = 1$, then $S = 1$

However, these informative events never occur together:

- When $X_1 = 1$, $X_2 = 0$
- When $X_2 = 1$, $X_1 = 0$

Ince's Criticism

Ince argues that this information is **not truly redundant**.

Each predictor provides information about the target, but that information appears on **different realizations**:

- X_1 has unique information on trials where $X_1 = 1$
- X_2 has unique information on trials where $X_2 = 1$

So although both predictors can tell us that $S = 1$, they do so on different trials.

What BROJA Does

BROJA preserves only the source-target marginals:

$$P(X_1, S)$$

and

$$P(X_2, S)$$

During the optimization, BROJA may change the joint relationship between X_1 and X_2.

In the optimized distribution, the informative events can become coupled, so both predictors appear to carry the same information on the same realizations.

As a result, BROJA classifies the information as **redundant**.

Main Lesson

The ReducedOR example shows that source-target marginals alone are not enough to decide whether information is redundant or unique.

The predictor-predictor relationship also matters:

$$P(X_1, X_2)$$

This relationship determines whether predictors are informative on the same trials or on different trials.

Key Takeaway

> Redundancy is not only about whether two predictors contain information about the same target value. It also depends on whether they provide that information on the same realizations.

💡 I am not sure Ince is right for that definition. It does seem that BROJA is conservative (which is good for us) but also I don't think $X_1 = 1$ as unique information is a good interpetation of this example. For me this example seems more like redundant and synergestic information.

Update: 06.07.2026:

We have decided that we are going with BROJA definition for unique information and we established enough examples to show that is the most logical definition for now - mostly because we it is blackwellian and more conservative under Gaussian assumption - meaning if it has to choose between zero unique information and above zero it will report zero unique information or equivalently if it shows above zero unique information it it is for sure above zero.

Now we progressed to try this definition on NSD data (fmri responses) and with real models. Recall that each subject in the data saw about 10,000 images and across subject 1,000 images were shared between them therefore we want to try the PID on the 1,000 shared images which it is a small number of sample in relation to the number of features.

The next challenge is **dimension reduction** because the fmri data and model's features are much bigger than the number of samples.

We have decided to work with next method for now and see where it goes:

Let T be the fmri data with number of sample n and dimension dt let X_1 and X_2 be the models features exacted (Now it is the extracted from the most predictable layer on the OTC found in Mayas_project using linear regression).

1. $\tilde{T} = PCA(T)$ where $\tilde{T}$ has a lower dimension than T

2. Regress X_1 and X_2 onto $\tilde{T}$ to have a lower dimension for them as well and now $d_1 = d_2 = d_t < n$ and on that we will preform PID.

The challenge in dimension reduction on fmri:

In order to choose the right number of components we want a PCA alogrithm that finds the most prediction PCs that explain the most of the variance but most of the algorithms are sensetive to noise and can overfit. For that we have to do cross-validation for PCA but this is tricky as well. Why? because regular cross validation like in linear regression does not fit in to that case.

see:

https://link.springer.com/article/10.1007/s00216-007-1790-1

or Alex's blog:

https://alexhwilliams.info/itsneuronalblog/2018/02/26/crossval/

They both explain on why regular cross validation will overfit with doing it on PCA.

Therefore there are 3 PCA - cross validation algorithms:

Let the centered data matrix be

$$X \in \mathbb{R}^{n \times d},$$

where

$$n = \text{number of samples}, \qquad d = \text{number of features}.$$

We test a candidate number of PCA components

$$f.$$

1. **Cross-validation by Eigenvector**
2. **EM-EK**
3. **EM-WOLD - Implemented in R by this package:** https://www.rdocumentation.org/packages/missMDA/versions/1.21/topics/estim_ncpPCA

NOTICE: When the correlation between the features is high the algorithms having a hard time.

Post - Eigen PID paper:

We were able to prove that minimum of union information or the unique information defined by BROJA can be derived in a closed formula defined thanks to the generalized eigenvalues between $COV(M|X)$ and $COV(M|Y)$. As a result we are able tom improve the computation time of BROJA unique information by more than 10^2.\

For more details see the arxiv paper - EIGEN-PID and the GITHUB.

We work on it all the way from August 2026 up to October 2026 and submitted it to AISTATS. Now we are back to our paper to check PID between to DNN models and FMRI data.

Plan for near future:

Now we need to think of a plan, How we want to write our paper, what sections should be? What models will we use? How do we divide them into groups? What second level analysis can we do?